



CAD Trust Data Schema v2.0.2

Schema Mapping Reference

Climate Action Data Trust
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This page is the developer reference for the CAD Trust Data Model v2.0.2 machine-readable schema. It covers every table, every field, all picklist values, validation rules, and migration notes from v1.

Schema file: `cadtrust-schema-v2.0.2.json`

Source: [Data Dictionary v2.0.2 \(Excel\)](#)

ERD: [Live Entity Relationship Diagram](#)

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What This Schema Is For

Registries connecting to CAD Trust v2.0 need to map their internal data to the CAD Trust data model before they can broadcast to the network. This schema makes that process machine-assisted rather than manual.

Use it to:

- **Drive ETL mapping tools** — load the schema, iterate over tables and fields, auto-suggest registry field → CAD Trust field mappings
- **Validate data before submission** — check required fields, enum values, date formats, and foreign key consistency without running a full Chia node
- **Build a sandbox validator** — accept registry data via API, run validation against this schema, return structured error reports
- **Generate documentation** — auto-render field reference pages, API docs, and integration guides from the schema
- **Power the AI mapping tool** — feed as context to an LLM to suggest mappings from a registry's own schema

Schema Structure

The JSON file has six top-level sections:

Section	Purpose
<code>\$schema, \$id, title, version</code>	Standard JSON Schema metadata
<code>changelog</code>	What changed from v1 to v2.0 — new tables, restructured tables, deprecated fields
<code>picklists</code>	All allowed enum values for every picklist field, by field name
<code>tables</code>	Full field definitions for all 15 tables
<code>validationRules</code>	Required fields for data activation, conditional requirements, enum notes
<code>migrationNotes</code>	Step-by-step guidance for v1 → v2 migration per changed concept

Tables Overview

15 tables total. 6 are new additions compared to Data Model Version 1.

Table	Status	Description	Parent FK
project	Existing	Core project record. All project-level tables reference this.	—
methodology	NEW v2.0	Standalone methodology registry with versioning	—
project_methodology	NEW v2.0	Junction: project ↔ methodology (many-to-many)	project, methodology

Table	Status	Description	Parent FK
location	Enhanced	Project geographic location. New: location_map_type, location_map_file_link	project
validation	Restructured	Validation event(s) for a project. Now separate from verification.	project
verification	Restructured	Periodic verification events. Links to validation. Referenced by issuance.	project, validation
issuance	Enhanced	Unit issuance event. Now requires FK to project_methodology.	verification, project_methodology, location
unit	Enhanced	Credit unit block. New: retirement detail, beneficiary, LEI, ITMOs reference ID.	issuance
stakeholder	NEW v2.0	Named project participants (owner, developer, consultant)	project
label	NEW v2.0	Credit eligibility labels (CORSIA, CCP, ITMO, etc.)	—
unit_label	NEW v2.0	Junction: unit ↔ label (many-to-many)	unit, label
co_benefit	Existing	SDG co-benefits linked to a project	project
estimation	Existing	Expected credit volumes for a future crediting period	project
rating	Existing	Independent project ratings (BeZero, CDP, CCQI, etc.)	project
program	NEW v2.0	Programme or activity grouping. Referenced from project.	—

AEF tables (aef_t1_submission, aef_t2_authorizations, aef_t3_actions, aef_t4_holdings, aef_t5_authorized_entities) are defined in the full schema JSON. They support Article 6 UNFCCC reporting and are only required for registries submitting AEF reports. See the schema JSON for full field definitions.

Field Definition Format

Every field in `tables.<table>.fields` follows this structure:

```
"unit_vintage_year": {
  "column": "unit_vintage_year",
  "apiField": "unitVintageYear",
  "type": "Year",
  "required": true,
  "source": "User Input",
  "description": "Year the units were awarded (vintage).",
  "aefMapping": "Vintage",
  "example": "2023"
}
```

Property	Meaning
column	Database column name (snake_case) — use in SQL/ETL
apiField	API field name (camelCase) — use in JSON payloads
type	Data type: String, Text, GUID, Date, Timestamp, Decimal, Year, URL, Picklist, Boolean
required	true = must be present for a valid record. false = optional
source	User Input = registry provides this. Inferred (system generated) = auto-generated by CAD Trust. API Generated = set by the API on create/update
picklist	Key into schema.picklists for allowed values (only on picklist fields)
foreignKey	table.field reference (only on FK fields)
aefMapping	Corresponding AEF report field name (only on fields used in Article 6 reporting)
description	Plain-English field description
example	Sample value

Validation Rules

Required fields for data activation

These 30 fields are the minimum a registry must populate for their data to appear in the CAD Trust Data Dashboard. The validator checks the required data fields as below

Entity Required	Fields
project	project_registry_name, project_id, project_name, project_link, project_sector, project_type, project_status, project_status_date, project_unit_metric
location	location_country
validation	validation_id, validation_type, validation_body
verification	verification_id, verification_body
issuance	issuance_id, cad_trust_verification_id, cad_trust_project_methodology_id, cad_trust_location_id
unit	unit_serial_id, unit_start_block, unit_end_block, unit_count, unit_type, unit_vintage_year, unit_status, unit_status_reason, unit_status_date, unit_metric
Methodology	methodology_code, methodology_name

Conditional requirements

Condition	Required field
unit.unit_status = Retired	unit.unit_retirement_detail (recommended)
unit.unit_status = Retired	unit.unit_retirement_beneficiary (recommended)
Unit has Article 6 authorization	unit.unit_itmos_reference_id
Unit carries eligibility labels	unit_label records required
Registry submitting AEF report	All aef_t1 through aef_t5 tables

Common enum mistakes

Wrong (v1)	Correct (v2.0)	Field
Active	Held	unit_status
Avoidance	Avoidance - nature or Avoidance - technical	unit_type
Single validation record covering both events	Separate validation • verification records	—
Methodology string on project	methodology record + project_methodology junction	—
No cad_trust_project_methodology_id on issuance	Required FK to project_methodology	issuance

Migration Guide: v1 to v2.0

1. methodology

v1: The methodology was stored as a single free-text string on the project record (`project.methodology`).

v2.0: Methodology is now a standalone record. Create one methodology record for each unique methodology in your data, then link it to the relevant project(s) using the `project_methodology_junction` table.

Migration steps:

For each unique methodology in your v1 data, create a corresponding methodology record:

```
INSERT INTO methodology (methodology_code, methodology_name, methodology_version)
VALUES ('VM0007', 'REDD+ Methodology', '1.6');
```

Then link the new methodology record to each project that uses it:

```
INSERT INTO project_methodology (cad_trust_project_id, cad_trust_methodology_id)
VALUES (<project_id>, <new_methodology_id>);
```

2. Validation and Verification

v1: Single combined record.

v2.0: Create one validation record and one verification record, optionally linked via `cad_trust_validation_id`.

3. Issuance, new required Foreign Key

Every issuance record now requires `cad_trust_project_methodology_id`. Backfill this by creating `project_methodology` records first, then updating issuance records.

4. unit_type enum values

v1 value	v2.0 mapping
Avoidance	→ Avoidance - nature or Avoidance - technical (registry to determine)
Reduction	→ Reduction - nature or Reduction - technical
Removal	→ Removal - nature or Removal - technical
Active (unit_status)	→ Held

5. Unit Labels (New Structure in V2)

If your registry tracks credit eligibility (CORSIA, CCP, ICVCM, etc.), create label records and link to units via `unit_label`.

6. Program (New Structure in V2)

If your registry previously stored a programme name as a free-text string on project, migrate to a program record and reference via `cad_trust_program_id` on the project.

Download

The full machine-readable schema JSON is available for download and programmatic use:

- **File:** `cadtrust-schema-v2.0.2.json`
- **Format:** JSON Schema
- **Size:** ~60 KB
- **Contents:** 15 tables · 140 fields · Full picklists · Validation rules · Migration notes

To use in a validator, ETL tool, or AI mapping assistant, load the JSON and reference `schema.tables`, `schema.picklists`, and `schema.validationRules`.